CASE STUDY / INTEGRATED PLANT INFRASTRUCTURE

Connected operations. Clear communication.

Element Chemilink Pvt. Ltd.
Ankleshwar | Gujarat

Networking • Matrix EPABX • Hikvision CCTV
Time-attendance solution design

One design. Distinct systems. Clear ownership.

Historical supply / installation scope and detailed integration drawings for customer understanding.



The engagement | A coordinated infrastructure solution

Element Chemilink requires practical connectivity, dependable internal telephony, recorded CCTV visibility and a clear attendance process. The supplied 2019 records establish computer, EPABX and CCTV procurement and a separate cabling quotation. D-TECH's integrated design connects these service needs while preserving the different signal and ownership paths.

System	Source-backed baseline	Status in this document
EPABX	PO ECPL/CAPEX/18-19/004, 11 March 2019: Matrix ETERNITY PE6SP, CO2+SLT6 card, 4 x SLT8, 1 x SLT4, 30-pair Krone junction box; installation / one-time training specified.	Ordered hardware with reference commissioning design.
CCTV	PO ECPL/CAPEX/18-19/005, 11 March 2019: 10 x 2 MP cameras, 16-channel DVR, camera SMPS, BNC / power connectors, 4 TB Seagate surveillance disk; installation rates specified.	Ordered analogue-HD camera / DVR path; no PoE-camera topology assumed.
NETWORK / VOICE CABLING	Quotation ERP-9854/5.00, 1 May 2019: 100 m 10-pair cable, 3 Krone boxes, 10 RJ45 IOs / back boxes / faceplates, casing and conduit; some quantities are "as actual".	Quoted passive scope; active LAN equipment / final lengths need approval.
COMPUTERS / ATTENDANCE	POs 4421000037 and 4421000039: three computer-set lines, one CPU-unit line and one separately listed HP M1005 MFP. No attendance terminal / software schedule supplied.	Historical endpoint supply; time attendance is user-requested solution design.

Reading guide: 03 Overall design | 04 LAN | 05 EPABX | 06 CCTV | 07 Attendance | 08 Deployment | 09 Acceptance and customer support

Overall architecture | Four services, separate signal paths

USER / ADMIN WORKSTATIONS

Approved LAN connection for business services and authorised recorder / attendance administration.

PROPOSED MANAGED LAN

Copper outlets → patching → managed switch → approved routing / firewall. Active equipment is not identified in the supplied scope.

TIME ATTENDANCE

Selected terminal → local buffer → approved application / HR review; device and software to be finalised.

ANALOGUE CCTV

Camera video → BNC / coax path → DVR. Camera DC power follows a separate SMPS distribution path.

16-CHANNEL DVR

10 cameras ordered; 4 TB disk ordered. Optional approved DVR LAN access for security / administrator users.

MATRIX EPABX

CO trunk / SLT cards → Krone → copper pairs → analogue handsets. Voice pairs are separate from LAN switching.

Solid lines show service paths; dashed line shows proposed authorised DVR LAN management. The PBX analogue voice circuit is not an Ethernet endpoint path.

No common single cable is assumed to carry camera DC power, analogue video, analogue voice and Ethernet without the appropriate engineered interface.

LAN design | Structured endpoints and controlled access

BUSINESS ENDPOINTS

Computer / printer outlets; terminal and recorder management only where approved.

RACK / MANAGED SWITCH

Labelled patch leads and port register; active switch selection, uplinks, power and rack inventory require site approval.

APPROVED SERVICE PATH

Customer router / firewall policy; no direct internet exposure of DVR or attendance administration.

Configuration detail	Implementation method
PASSIVE VERIFICATION	Map quoted IOs / faceplates and cable routes; verify RJ45 use for voice versus Ethernet. Treat “as actual” cable lines as measured work, not confirmed 305 m supply.
SEGMENTATION	Where supported and approved, separate business, recorder management, attendance and network administration. Assign actual VLAN IDs / subnets after survey.
PORT / ADDRESS PLAN	Record wall outlet, rack port, device, address and owner. Restrict management to approved users; configure time synchronisation and change records.
ACCEPTANCE	Test cable continuity / wiremap and suitable performance; verify permitted access, blocked paths and configuration backup. Record switch model and capacity.

Network segmentation is a proposed integration design. No managed-switch, firewall or Wi-Fi equipment quantity is established by the supplied documents.

EPABX design | Matrix platform and copper distribution

CO / EXTERNAL LINES

Ordered CO2+SLT6 card provides the labelled CO interface; actual carrier lines and service availability require confirmation.

MATRIX ETERNITY PE6SP

1 CO2+SLT6 + 4 SLT8 + 1 SLT4 cards. Label-based arithmetic: 2 CO and 42 SLT ports; verify approved chassis / card compatibility.

KRONE → SLT HANDSETS

30-pair junction box in PO; 3 x 10-pair boxes in quote. Final installed port count and pair allocation must be reconciled.

Micro-level task	Commissioning and customer understanding
EXTENSION PLAN	Create department-wise extension / location list; assign available SLT ports to actual handsets. Port arithmetic is not proof of 42 installed telephones.
PAIR / TERMINATION MAP	Label chassis card / port → Krone pair → cable pair → outlet → extension. Verify enough pairs for active extensions and identify spare pairs.
CALL POLICY & TRAINING	Approve incoming destinations, day / night routing and outbound restrictions; test internal calls, transfer and authorised external calls; provide the ordered one-time training.

No SIP trunk, IP-phone licence, call recording or mobile integration is assumed. Existing carrier, handset, earthing and power-protection details require survey.

CCTV design | Separate video, power and recording paths

10 x 2 MP CAMERAS

DS-2CE16D0T-IRPF in PO. Verify actual camera housing: PO says dome, while the OEM model datasheet identifies a bullet camera.

BNC VIDEO → 16CH DVR

Hikvision DS-7216HQHI-K2; 32 BNC pins ordered. Map each camera / video run to a DVR channel and scene.

4 TB SURVEILLANCE DISK

One Seagate 4 TB HDD. Configure supported recording parameters and validate disk health / usable capacity.

SEPARATE DC POWER

16-channel camera SMPS; 16 power pins ordered. Verify voltage, load and voltage drop at each camera.

SECURITY USER

Approved live view, playback and export. Apply named permissions and record time / recording schedules.

RETENTION VALIDATION

Illustration: 10 cameras × 2 Mbps ≈ 216 GB/day. Raw 4 TB gives about 18.5 days before overhead; actual settings govern.

Ten ordered cameras use part of a 16-channel recorder’s nameplate capacity; expansion needs compatible cameras, power, cabling, storage and performance checks.

Time attendance | Capture, reconcile and approve

01 | ENROL & ASSIGN

Create approved employee ID, department, shift and credential mapping; restrict administrator access.

02 | CAPTURE PUNCH

Selected terminal records employee, device and timestamp; define in / out method and local event buffering.

03 | SYNC EVENTS

Synchronise to the chosen application using a supported interface; track last event and detect duplicates.

04 | APPLY SHIFT RULES

Calculate attendance using agreed shifts, overnight rules, late / early policy and missing-punch handling.

05 | HR REVIEW

Authorised HR reviews exceptions; record correction reason, approver and audit history.

06 | REPORT / EXPORT

Issue approved attendance reports; payroll export only after format, mapping and responsibility are agreed.

Select credential method, terminal count, user capacity, application hosting and retention with the client. Attendance capture does not itself authorise door access.

Test offline buffering / resync, duplicate and missing events, time zone and overnight shift handling before acceptance. No payroll or ERP API is assumed.

Deployment | From survey to customer handover

01 | SURVEY

Verify premises, endpoint locations, cable routes, existing power, PBX handsets, camera views and attendance policy.

02 | APPROVE DESIGN

Freeze equipment / quantities, pair and port plans, user roles, recording settings, attendance rules and work windows.

03 | INSTALL & LABEL

Install approved containment, cable and termination; label rack ports, voice pairs, DVR channels and devices.

04 | CONFIGURE

Commission approved LAN, PBX call plan, DVR recording and attendance application; preserve configuration records.

05 | TEST & TRAIN

Run service-specific tests with client owners; provide PBX training and agreed operator / HR guidance.

06 | HAND OVER

Submit registers, drawings, backups, test sheets, entitlements and open issues; obtain customer sign-off.

EPABX / CCTV installation terms appear in the historical orders; broader LAN and attendance implementation require the final approved scope.

Acceptance and support | Clear evidence, clear ownership

System / owner	Acceptance evidence
NETWORK / CUSTOMER IT	Outlet and device mapping; cable tests; permitted / blocked access tests; approved switch / IP register and configuration backup.
EPABX / ADMINISTRATION	Card / port / pair register; internal calls, transfer and approved trunk routing; extension restrictions; one-time training record.
CCTV / SECURITY	All ordered camera views; power / video health; correct timestamp; recording and playback; export; measured available retention.
ATTENDANCE / HR	Employee / shift mapping; offline / resync test; duplicate and missing-punch exceptions; approved correction and report / export tests.
D-TECH / SERVICE COORDINATION	Ticket → assessment → customer approval → resolution / specialist escalation → validation → closure. Record actions, evidence and acceptance.

CUSTOMER / LOCATIONS

Element Chemilink Pvt. Ltd., Ankleshwar
Office: Plot 2601/11, GIDC Industrial Estate.
Unit-II: Plots 313/6, 7 & 8, GIDC, Ankleshwar 393002.

D-TECH | CUSTOMER SUPPORT

support.dtechindia.com | Open support portal

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Basis: 4 supplied 2019 POs, quotation ERP-9854/5.00, client images and the requested LAN / attendance scope. Drawings are reference designs, not as-built records.

OEM reference: Hikvision DS-2CE16D0T-IRPF datasheet | Camera-model identification